



ELA: English Language Assistant

An AI-Assisted Framework for Authentic English Comprehension Using Recursive Linguistic Annotation

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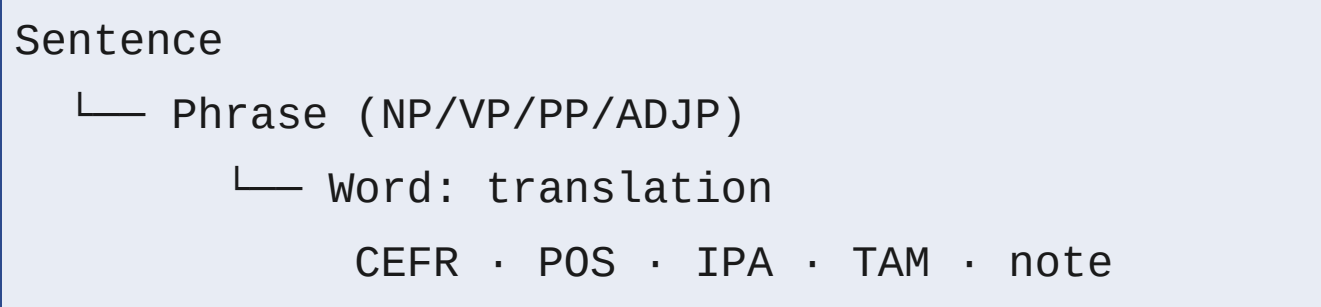
Supervisor Semester 2 (Implementation Phase): Dr. Nasir Ahmad

Department of Computer Science · Faculty of Engineering and Science · Munster Technological University Cork · May 2025

Problem & Motivation

Mainstream language learning platforms present **pre-simplified content**, stripping the grammatical structures and vocabulary range that define authentic English. Learners are shielded from complexity rather than guided through it. ELA annotates authentic text and speech **in place** — **structure intact** — using rule-based NLP and fine-tuned ML models. Each sentence is decomposed into a **Recursive Linguistic Element (RLE)** tree enriched with CEFR level, tense–aspect–mood (TAM) tags, IPA, and bilingual notes — processed fully offline, client-side, without cloud dependency.

The RLE Data Model



Language-agnostic · EN → RU via M2M100 (offline).

Analysis Pipeline



Research Journey: What We Tried

CEFR CLASSIFICATION	NOTE GENERATION
DeBERTa underperformed BERT same XGBoost ✓ 78.3% Whisper ✓ WER 8.2%	GPT-4o east/cloud T5 raw no-generalise T5+Templates ✓ ROUGE-L 0.81 GPT-4o opt-in M2M100 ✓ offline

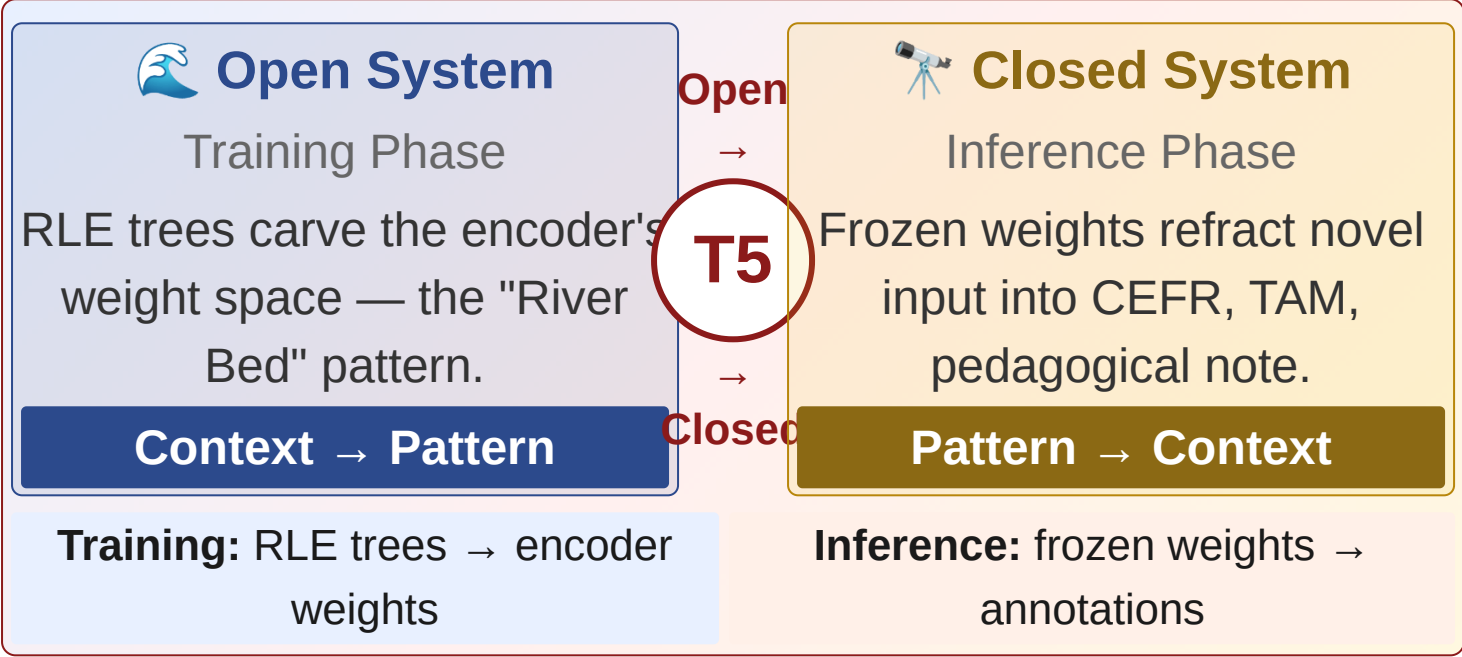
Context–Pattern Interaction

"In open systems, context forms patterns. In closed systems, patterns form context."
— Discovered empirically during T5 fine-tuning

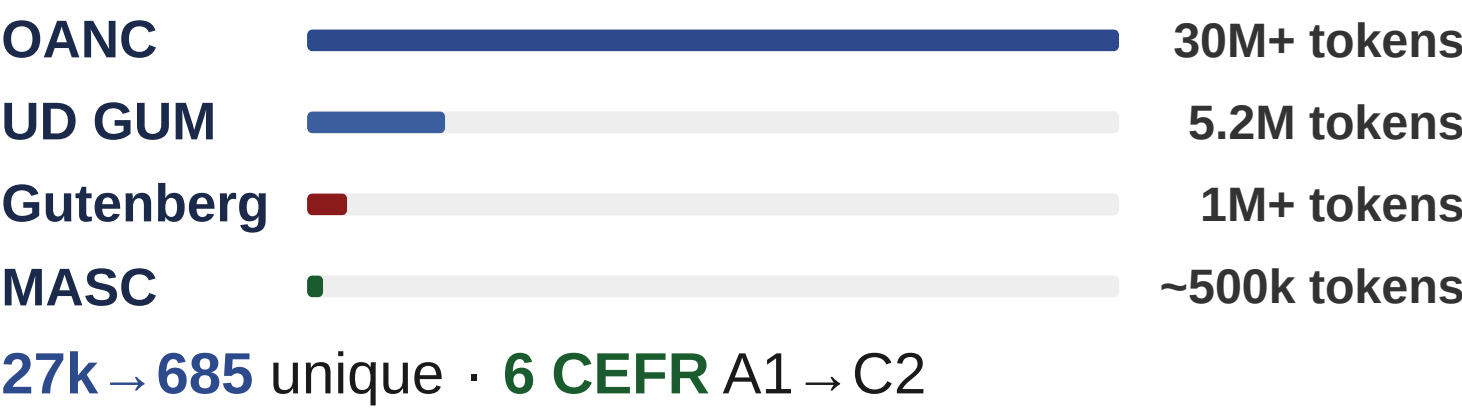
T5 on raw notation **failed**. Structural templates — **lexically invariant** — were the breakthrough:

✗ "gave him" → [VP gave][NP him]
✓ [V-PAST][IOBJ] → [TAM:past-simple]
[VOICE:active]

The Duality Loop Diagram



Training Data & Corpus Scale



Annotation Backend Comparison

Approach	Quality	Cost	Offline
GPT-4o	High	\$\$\$	✗
Rules only	Medium	Free	✓
T5 fine-tuned	High	Free	✓
XGBoost CEFR	78.3%	Free	✓

System Validation

- **O1:** spaCy + TAM → schema-valid RLE ✓
- **O2:** CEFR 78.3%; T5 ROUGE-L 0.81 ✓
- **O3:** Visualizer + JSON/CSV export ✓
- **O4:** Android PWA, fully client-side ✓

Performance Metrics

78.3% CEFR Accuracy XGBoost ensemble	0.81 ROUGE-L T5 fine-tuned	8.2% ASR WER Whisper base
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Conclusions

- AI annotation makes authentic English **accessible without simplification**.
- The **Context ↔ Pattern duality** generalises to any open → closed system; T5 placeholders are **lexically invariant**.

Future Directions

- **Tauri 2 Desktop App** — native .deb/Windows with bundled ML models.
- **Translation Quality Gates** — ChrF scorer, M2M100 review queue.
- **M2M100 LoRA Fine-Tuning** — domain-specific EN → RU refinement.
- **Real-time Streaming ASR** — Whisper streaming for live practice.



Source Code

github.com/welsol21/FYP_LLM



Live Demo

www.el-a.uk

References

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Acknowledgements

The author sincerely thanks **Dr. Nasir Ahmad** (Supervisor Semester 2, Implementation Phase) and **Dr. Alex Vakaloudis** (Supervisor Semester 1, Research Phase) for their invaluable guidance throughout this project. Thanks to the Department of Computer Science at MTU Cork for the academic environment that made this work possible. Inspired by *Eleonora Rastvorova*.